# Practical 3 - Student Rating Dataset - Statistical Analysis

**Problem -** Using the Student Rating Dataset and answer the following:

1. Calculate the percentage of visible minorities are tenure professors. Will you say that tenure status differed if the teacher was a visible minority?
2. Does average age differ by tenure? Produce the means and standard deviations for both tenured and untenured professors.
3. Which graph works better and plot that graph for the age variable?
4. Define the difference between pyplot.bar and pyplot.barh. Also plot the graph for gender variables.
5. What is the Median evaluation score for tenured Professors?

**Solution -** Here is the complete code.

**Conclusion -** The analysis of the TeachingRatings dataset shows differences in age between tenured and untenured professors. The age distribution is best represented using a histogram, while gender is effectively displayed using a bar graph. Overall, the dataset provides useful insights into tenure, age, gender, visible-minority status, and teaching evaluation scores of professors.

## Import Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
```

## Load Data

```
url = "https://raw.githubusercontent.com/vincentarelbundock/Rdatasets/master/csv/AER/TeachingRatings.csv"

df = pd.read_csv(url)
```

```
df.head()
```

| | rownames | minority | age | gender | credits | beauty | eval | division | native | tenure | students | allstudents | prof |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | yes | 36 | female | more | 0.289916 | 4.3 | upper | yes | yes | 24 | 43 | 1 |
| **1** | 2 | no | 59 | male | more | -0.737732 | 4.5 | upper | yes | yes | 17 | 20 | 2 |
| **2** | 3 | no | 51 | male | more | -0.571984 | 3.7 | upper | yes | yes | 55 | 55 | 3 |
| **3** | 4 | no | 40 | female | more | -0.677963 | 4.3 | upper | yes | yes | 40 | 46 | 4 |
| **4** | 5 | no | 31 | female | more | 1.509794 | 4.4 | upper | yes | yes | 42 | 48 | 5 |

Next steps: Generate code with df | New interactive sheet

Ques 1. Calculate the percentage of visible minorities are tenure professors. Will you say that tenure status differed if the teacher was a visible minority?

```
minority_tenured = df[
    (df["minority"] == "yes") &
    (df["tenure"] == "yes")
].shape[0]

total_minority = df[df["minority"] == "yes"].shape[0]
```

```
percentage = (minority_tenured / total_minority) * 100

print("Percentage:", percentage, "%")
```

```
Percentage: 84.375 %
```

**Interpretation**: The percentage tells us what proportion of professors classified as visible minorities are on the tenure track. The dataset defines minority as whether the instructor belongs to a minority group and tenure as whether the instructor is on the tenure track.

**Ques 2.** Does average age differ by tenure? Produce the means and standard deviations for both tenured and untenured professors.

```
df.groupby("tenure")["age"].agg(["mean", "std"])
```

| tenure | mean | std |
|---|---|---|
| no | 50.186275 | 6.946372 |
| yes | 47.850416 | 10.420056 |

**Interpretations-** Compare the two means. The group with the larger mean is older on average, while the standard deviation shows how much the ages vary within each tenure group. The overall dataset has a mean age of 48.365 years and SD of 9.803 years.

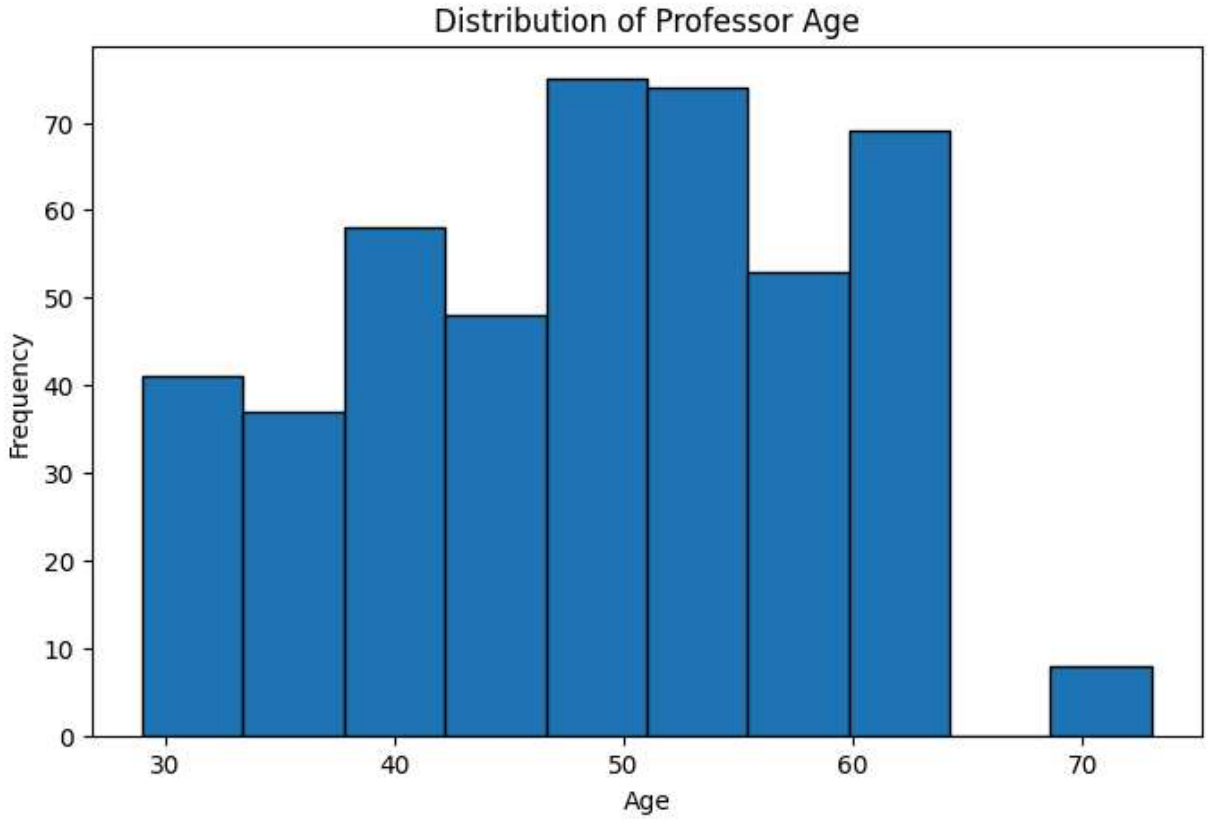
**Ques 3.** Which graph works better and plot that graph for the age variable?

```
plt.figure(figsize=(8,5))

plt.hist(df["age"], bins=10, edgecolor="black")

plt.xlabel("Age")
plt.ylabel("Frequency")
plt.title("Distribution of Professor Age")

plt.show()
```




**Interpretations-** A histogram is appropriate because age is a numerical/continuous variable.The histogram shows how professors' ages are distributed and which age ranges occur most frequently.

**Ques 4.** Define the difference between pyplot.bar and pyplot.barh. Also plot the graph for gender variables.

```
gender_counts = df["gender"].value_counts()

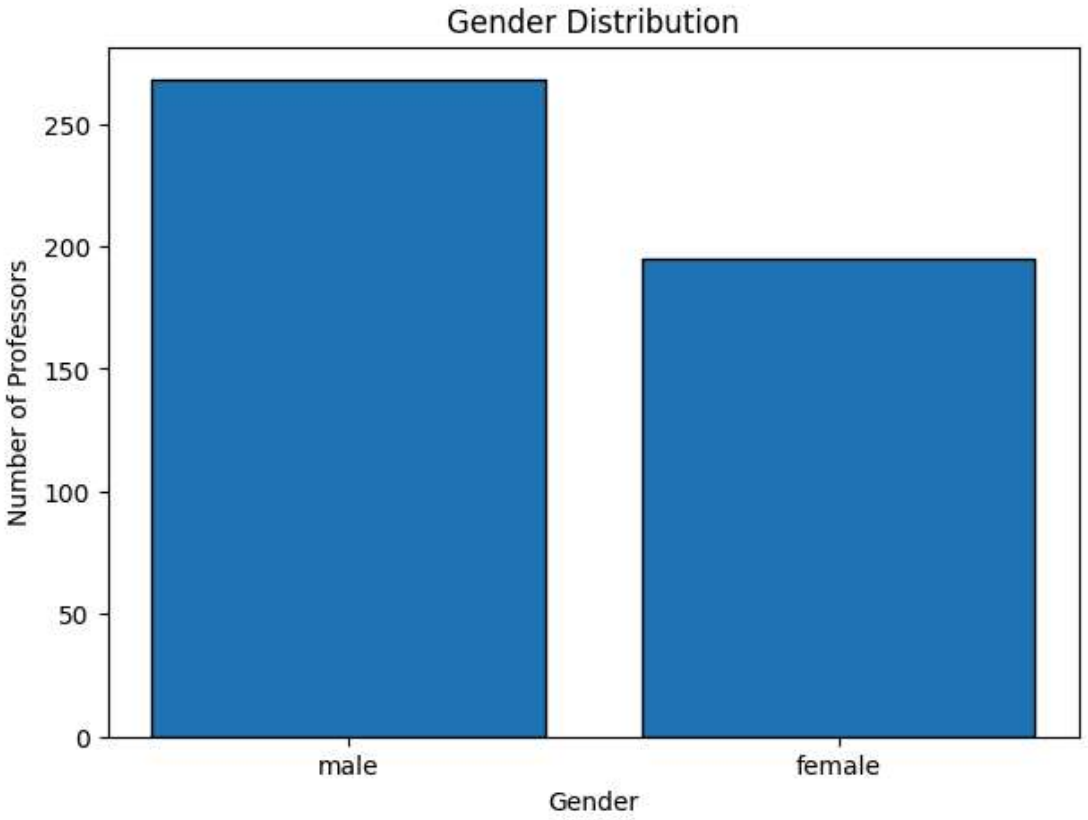
plt.figure(figsize=(7,5))

plt.bar(gender_counts.index, gender_counts.values, edgecolor="black")

plt.xlabel("Gender")
plt.ylabel("Number of Professors")
plt.title("Gender Distribution")
```

```
plt.show()

print(gender_counts)
```




```
gender
male      268
female    195
Name: count, dtype: int64
```

**Interpretations-**

pyplot.bar() - vertical bars

pyplot.barh() - horizontal bars

The graph shows that there are more male professors than female professors in the dataset.

**Ques 5.** What is the Median evaluation score for tenured Professors?

```
median_eval = df[df["tenure"] == "yes"]["eval"].median()

print("Median evaluation score for tenured professors:", median_eval)
```

```
Median evaluation score for tenured professors: 4.0
```

**Interpretations - **

The median represents the middle evaluation score among tenured professors. The overall dataset has a median evaluation score of 4.0

```
Start coding or generate with AI.
```